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United States Patent

12402692

Kind Code

B2

Date of Patent

September 02, 2025

Inventor(s)

Campos, II; Fidencio et al.

Sole structure with midsole protrusions and arced profile for forward momentum

Abstract

A footwear sole structure includes a midsole having a ground-facing surface with a forefoot region, a midfoot region, and a heel region. The midsole defines downwardly-extending protrusions at the ground-facing surface distributed over the midfoot region and the heel region, each of the downwardly-extending protrusions having a convex outer surface. A height of the midsole is greatest at the midfoot region. The sole structure may also include an outsole covering at least a portion of the ground-facing surface of the midsole, and a height of the sole structure may be greatest at the midfoot region.

Inventors: Campos, II; Fidencio (Dallas, OR), Dimoff; Karen S. (Portland, OR), Fitchett; Derek A. (Beaverton, OR), Harrington; Emily (Beaverton, OR), Helseth; Joseph L. (Plano, TX), McLachlan; Oliver (Beaverton, OR), Runkle; Colin (Portland, OR)

Applicant: NIKE, Inc. (Beaverton, OR)

Family ID: 77227123

Assignee: NIKE, Inc. (Beaverton, OR)

Appl. No.: 18/307179

Filed: April 26, 2023

Prior Publication Data

Document Identifier

US 20230255314 A1

Publication Date

Aug. 17, 2023

Related U.S. Application Data

Publication Classification

Int. Cl.: A43B13/22 (20060101); A43B13/14 (20060101); A43B13/18 (20060101)

U.S. Cl.:

CPC A43B13/223 (20130101); A43B13/145 (20130101); A43B13/184 (20130101);

Field of Classification Search

CPC: A43B (13/223); A43B (13/125); A43B (13/184)

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Primary Examiner: Lynch; Megan E

Attorney, Agent or Firm: Quinn IP Law

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of U.S. Nonprovisional application Ser. No. 17/372,733, filed Jul. 12, 2021, now issued as U.S. Pat. No. 11,666,119, issued Jun. 6, 2023, which claims the benefit of priority to U.S. Provisional Application No. 63/051,110, filed Jul. 13, 2020, and each which is hereby incorporated by reference in its entirety.

TECHNICAL FIELD

(1) The present disclosure generally relates to a sole structure for an article of footwear including a midsole with downwardly-extending protrusions.

BACKGROUND

(2) Footwear typically includes a sole assembly configured to be located under a wearer's foot to space the foot away from the ground. Sole assemblies in athletic footwear are configured to provide desired cushioning, motion control, and resiliency and are often composed of multiple components of different materials in order to meet durability, stability, and cushioning goals. For example, some components may have high energy return and elastic resiliency under compressive loading, while other components may have less elastic resiliency but greater abrasion resistance. Footwear manufacturers strive to design and assemble the various components to enable each to achieve its functionality.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) The drawings described herein are for illustrative purposes only, are schematic in nature, and are intended to be exemplary rather than to limit the scope of the disclosure.
- (2) FIG. 1 is a bottom view of a sole structure for an article of footwear including a midsole and an outsole.
- (3) FIG. 2 is a perspective bottom and lateral side view of the midsole without the outsole.
- (4) FIG. 3 is a lateral side view of the sole structure.
- (5) FIG. 4 is a medial side view of the sole structure.
- (6) FIG. 5 is a cross-sectional view of the sole structure taken at lines 5-5 in FIG. 15.
- (7) FIG. 6 is a cross-sectional view of the sole structure taken at lines 6-6 in FIG. 1.
- (8) FIG. 7 is a schematic illustration of the sole structure in cross-sectional view worn on a foot shown in phantom during ground contact of a heel region such as during a heel strike of a forward

stride.

(9) FIG. 8 is a schematic illustration of the sole structure of FIG. 7 during ground contact of a midfoot region during the forward stride.

(10) FIG. 9 is a schematic illustration of the sole structure of FIG. 7 during ground contact of a forefoot region during a forward stride.

(11) FIG. 10 is a cross-sectional illustration of the sole structure taken at lines 10-10 in FIG. 15.

(12) FIG. 11 is a cross-sectional illustration of the sole structure taken at lines 11-11 in FIG. 15.

(13) FIG. 12 is a cross-sectional illustration of the sole structure taken at lines 12-12 in FIG. 15.

(14) FIG. 13 is a front view of the sole structure.

(15) FIG. 14 is a rear view of the sole structure.

(16) FIG. 15 is a top view of the sole structure.

(17) FIG. 16 is a bottom view of another embodiment of a sole structure for an article of footwear including a midsole and an outsole.

(18) FIG. 17 is a perspective view of the sole structure of FIG. 16.

(19) FIG. 18 is a perspective view of the midsole of the sole structure of FIG. 16.

(20) FIG. 19 is a bottom view of the midsole of the sole structure of FIG. 16.

(21) FIG. 20 is a bottom view of the outsole of the sole structure of FIG. 16.

(22) FIG. 21 is a perspective view of the outsole of the sole structure of FIG. 16.

(23) FIG. 22 is a top view of the outsole of the sole structure of FIG. 16.

(24) FIG. 23 is a cross-sectional view of the outsole of FIG. 16 taken at lines 23-23 in FIG. 22.

(25) FIG. 24 is a cross-sectional view of the sole structure of FIG. 16 taken at lines 24-24 in FIG. 16.

(26) FIG. 25 is a top view of the sole structure of FIG. 16.

(27) FIG. 26 is a front view of the sole structure of FIG. 16.

(28) FIG. 27 is a rear view of the sole structure of FIG. 16.

(29) FIG. 28 is a cross-sectional view of the sole structure of FIG. 16 taken at lines 28-28 in FIG. 25.

(30) FIG. 29 is a cross-sectional view of the sole structure of FIG. 16 taken at lines 29-29 in FIG. 25.

(31) FIG. 30 is a cross-sectional view of the sole structure of FIG. 16 taken at lines 30-30 in FIG. 25.

(32) FIG. 31 is a cross-sectional view of the sole structure of FIG. 16 taken at lines 31-31 in FIG. 25.

(33) FIG. 32 is a bottom view of another embodiment of a sole structure for an article of footwear including a midsole and an outsole.

(34) FIG. 33 is a medial side view of the sole structure of FIG. 32.

(35) FIG. 34 is a lateral side view of the sole structure of FIG. 32.

(36) FIG. 35 is a cross-sectional view of the sole structure of FIG. 32 taken at lines 35-35 in FIG. 32.

(37) FIG. 36 is a top view of the sole structure of FIG. 32.

(38) FIG. 37 is a front view of the sole structure of FIG. 32.

(39) FIG. 38 is a rear view of the sole structure of FIG. 32.

(40) FIG. 39 is a cross-sectional view of the sole structure of FIG. 32 taken at lines 39-39 in FIG. 36.

(41) FIG. 40 is a cross-sectional view of the sole structure of FIG. 32 taken at lines 40-40 in FIG. 36.

(42) FIG. 41 is a cross-sectional view of the sole structure of FIG. 32 taken at lines 41-41 in FIG. 36.

(43) FIG. 42 is a cross-sectional view of the sole structure of FIG. 32 taken at lines 42-42 in FIG. 36.

- (44) FIG. 43 is a bottom view of another embodiment of a sole structure for an article of footwear including a midsole and an outsole.
- (45) FIG. 44 is a medial side view of the sole structure of FIG. 43.
- (46) FIG. 45 is a lateral side view of the sole structure of FIG. 43.
- (47) FIG. 46 is a cross-sectional view of the sole structure of FIG. 43 taken at lines 46-46 in FIG. 43.
- (48) FIG. 47 is a top view of the sole structure of FIG. 43.
- (49) FIG. 48 is a front view of the sole structure of FIG. 43.
- (50) FIG. 49 is a rear view of the sole structure of FIG. 43.
- (51) FIG. 50 is a cross-sectional view of the sole structure of FIG. 43 taken at lines 50-50 in FIG. 47.
- (52) FIG. 51 is a cross-sectional view of the sole structure of FIG. 43 taken at lines 51-51 in FIG. 47.
- (53) FIG. 52 is a cross-sectional view of the sole structure of FIG. 43 taken at lines 52-52 in FIG. 47.
- (54) FIG. 53 is a cross-sectional view of the sole structure of FIG. 43 taken at lines 53-53 in FIG. 47.

DESCRIPTION

- (55) The present disclosure generally relates to a sole structure for an article of footwear having features that may be especially advantageous for a variety of forward paces, including walking, relatively slow running, and for leisure wear, and/or a combination of these activities. For example, the sole structure may include a midsole that promotes a soft landing upon impact, and encourages an efficient and relatively even forward momentum through a forward stride from heel impact, through the midfoot region, to toe-off from a supportive and relative stiff platform at the forefoot region. Walkers and slower runners may tend to land on the heel region more frequently than a faster performance runner, so a design that both mitigates heel impact and promotes a smooth, efficient transition from heel to toe is especially beneficial for these activities and paces.
- (56) More specifically, a sole structure for an article of footwear may include a midsole having a ground-facing surface with a forefoot region, a midfoot region, and a heel region. The midsole may define downwardly-extending protrusions at the ground-facing surface distributed over the midfoot region and the heel region. Each of the downwardly-extending protrusions may have a convex outer surface. A height of the midsole may be greatest at the midfoot region.
- (57) Additionally, a height of the midsole may be greater at the midfoot protrusions than at the heel protrusions, and greater at the midfoot protrusions than at the forefoot region. Such an embodiment enables a “rocker” functionality of the midsole. For example, the ground-facing surface of the midsole may curve upwardly from the midfoot region to a forward extent of the midsole and from the midfoot region to a rear extent of the midsole, establishing an arced profile of the midsole. With such a full-length, convex camber, only a relatively small area of the ground-facing surface is in contact with a horizontal ground plane at any time during a forward stride, and the rate of transitioning forward on the midsole is relatively constant in comparison to sole structures configured so that a large portion of the midfoot region comes into contact with the ground very abruptly upon transition from a heel region to a midfoot region, for example. This helps to avoid a “slapping” phenomena and associated foot fatigue that may occur with such sole structures configured so that a large portion of the midfoot region comes into contact with the ground very abruptly upon transition from a heel region to a midfoot region.
- (58) In an implementation, the ground-facing surface of the midsole may curve upwardly from the midfoot region to a forward extent of the midsole and from the midfoot region to a rear extent of the midsole, establishing an arced profile of the midsole.
- (59) In contrast to the midfoot region and heel region having the downwardly-extending protrusions, in one or more implementations, the ground-facing surface of the forefoot region may

be relatively flat. To provide a stable platform for toe-off from the forefoot region, downwardly-extending protrusions may be absent from at least a forward half of the forefoot region.

(60) The midsole may include a rear side wall that flares outward from an upper extent to a lower extent of the rear side wall at a rear of the heel region. This may help to steer the midsole into the forward rocking motion early in the forward stride. The compressibility of the protrusions mitigates impact to protect against muscle fatigue, while the arced profile promotes an efficient transition from heel strike to toe-off. Typically, it is difficult to achieve both of these goals, as increasing compressibility often decreases the efficiency of forward motion (e.g., the wearer may need to exert more energy to maintain forward momentum in a highly compressible, cushioned midsole without an arced profile than in one with an arced profile).

(61) In an implementation, the midsole may include a lateral side wall and a medial side wall each of which has an upper extent and a lower extent and each of which flares outward from the upper extent to the lower extent in the forefoot region. Accordingly, the forefoot region may be relatively flat and wide. The height of the midsole in the forefoot region should be sufficient to provide adequate cushioning, while the relative flatness of the foam (absence of the downwardly-extending protrusions) makes this region relatively stiff in comparison to other regions to provide support for an efficient toe-off.

(62) Additionally, the downwardly-extending protrusions may include forwardmost protrusions each having a front half and a rear half, and the convex outer surface may be steeper at the rear half than at the front half. The front half, being less steep, may more gradually extend into a relatively flat forefoot region of the ground-facing surface forward of the forwardmost protrusions (e.g., relatively flat in comparison to the midfoot and heel regions).

(63) For stability in the midfoot region given the convex shapes of the downwardly-extending protrusions, the downwardly-extending protrusions may include peripheral protrusions in the midfoot region that define a lateral side edge and a medial side edge of the ground-facing surface. The peripheral protrusions may be truncated at the lateral side edge and at the medial side edge such that peaks of the peripheral protrusions lie along the lateral side edge and the medial side edge. The peripheral protrusions provide widely spaced contact areas with the ground plane, increasing medial-lateral stability. Similar truncated peripheral protrusions may define a rear edge with peaks lying along the rear edge for stability upon heel impact.

(64) The midsole may be a one-piece foam body. For example, each of the downwardly-extending protrusions and the base from which they extend may be a single, unitary, one-piece component. For example, a foam material may be injection molded, compression molded, or otherwise manufactured as a foam body that is a one-piece component. In some examples, the foam material may comprise an EVA foam, such as a blend of EVA material or materials, for example.

(65) In an implementation, the sole structure may include an outsole covering at least a portion of the ground-facing surface of the midsole. A height of the sole structure may be greatest at the midfoot region. Stated differently, the height of the sole structure, including the height of both the midsole and the outsole, may be greatest at the midfoot region. Accordingly, like the midsole, the sole structure (including both the midsole and the outsole) also has an arced profile.

(66) In an implementation, the sole structure may include an outsole that has an outsole element secured to the ground-facing surface of the midsole in the forefoot region. For example, the outsole element may be a first outsole element, and the outsole may further include a second outsole element covering the ground-facing surface of the midsole in the heel region.

(67) In an implementation, the outsole may further include a third outsole element covering the ground-facing surface of the midsole in the midfoot region. The first outsole element and the third outsole element may be separated by a first gap extending from a medial edge to a lateral edge of the midsole. The midsole may include a first ridge that extends into and at least partially fills the first gap. Similarly, the second outsole element and the third outsole element may be separated by a second gap extending from the medial edge to the lateral edge of the midsole. The midsole may

include a second ridge that extends into and at least partially fills the second gap between the second outsole element and the third outsole element. In some configurations, the first ridge and the second ridge may be nonlinear.

(68) In implementations such as those in which the outsole includes discrete outsole elements separated from one another, the flexibility of the midsole may be less constrained by the outsole in comparison to an embodiment in which a one-piece outsole extends across most or all of the ground-facing surface.

(69) In an implementation, the outsole may be thicker at peaks of the downwardly-extending protrusions than at the ground-facing surface of the midsole between adjacent peaks of the downwardly-extending protrusions. The outsole may be a relatively durable material in comparison to the material of the midsole, and the thickness at the peaks aids in durability of the sole structure, slowing wear. The material of the midsole may be relatively more compressible than the material of the outsole, and the relative thinness of the outsole between the peaks may therefore enable greater movement and deformation of the downwardly-extending protrusions during compression in comparison to a thicker outsole between the peaks. Stated differently, an outsole with relatively thin areas between the peaks may constrain the midsole less than if the material between the peaks were thicker, allowing greater resilient deformation and related cushioning.

(70) In an implementation, the midsole may include a medial side wall having a lower medial side edge and a lateral side wall having a lower lateral side edge. The outsole may extend to and underlie the medial side edge and the lateral side edge, terminating at the medial side edge and the lateral side edge without extending onto the medial side wall and the lateral side wall. By not extending onto the medial side wall and the lateral side wall, the midsole is less constrained by the outsole and may resiliently deform to a greater degree, providing greater cushioning.

(71) In an implementation, the outsole may include a forefoot outsole element, a midfoot outsole element, and a heel outsole element. Each of the forefoot outsole element, the midfoot outsole element, and the heel outsole element may extend from a medial side wall to a lateral side wall of the midsole. A rear edge of the forefoot outsole element may be spaced apart from a forward edge of the midfoot outsole element defining a first gap between the rear edge of the forefoot outsole element and the forward edge of the midfoot outsole element. A rear edge of the midfoot outsole element may be spaced apart from a forward edge of the heel outsole element defining a second gap between the rear edge of the midfoot outsole element and the forward edge of the heel outsole element. Because the midsole may be more flexible than the outsole, the gaps allow greater movement of the midsole during dorsiflexion, for example, than if the outsole extended without gaps along a ground-facing surface of the midsole from a heel region to a forefoot region of the midsole.

(72) Still further, the rear edge of the forefoot outsole element may have an irregular shape, and the forward edge of the midfoot outsole element may have a complementary irregular shape that tracks the irregular shape of the rear edge of the forefoot outsole element.

(73) Similarly, the forward edge of the heel outsole element may have an irregular shape, and the rear edge of the midfoot outsole element may have a complementary irregular shape that tracks the irregular shape of the forward edge of the heel outsole element.

(74) By providing edges of adjacent outsole elements that are complementary and track one another, the competing goals of covering the ground-facing surface of the midsole with the outsole to increase durability and allowing flexibility and deformation of the midsole without excessive constraint by the outsole may both be achieved.

(75) In an implementation, the midsole may include a first ridge that extends into and at least partially fills the first gap, and the midsole may further include a second ridge that extends into and at least partially fills the second gap. The ridges of the midsole may thus define a portion of the ground-engaging surface.

(76) In one or more implementations, an outsole element may cover the ground-facing surface of

the midsole only in the forefoot region. Stated differently, the ground-facing surface of the midsole in the midfoot region and in the heel region may serve as the ground-engaging surface along with the outsole element in the forefoot region. In one such configuration, forwardmost protrusions of the downwardly-extending protrusions are rearward of a widest portion of the midsole in the forefoot region, and a majority of the outsole element is forward of the widest portion of the midsole in the forefoot region. Providing an outsole element in the forefoot region may enable increased durability and traction needed for toe-off. In some embodiments, the material of the midsole alone may provide sufficient durability and traction such that no outsole element is included.

(77) In a configuration, the sole structure may comprise a midsole having a ground-facing surface with a forefoot region, a midfoot region, and a heel region. The midsole may define downwardly-extending protrusions at the ground-facing surface distributed over the midfoot region and the heel region and absent from at least a forward half of the forefoot region. Each of the downwardly-extending protrusions may have a convex outer surface. The downwardly-extending protrusions may include midfoot protrusions in the midfoot region having widths in a transverse direction of the midsole greater than lengths in a longitudinal direction of the midsole. The downwardly-extending protrusions may also include heel protrusions in the heel region having width-to-length ratios less than width-to-length ratios of the midfoot protrusions. For example, the midfoot protrusions may be relatively oblong and the heel protrusions may be relatively round.

(78) To promote resilient deformation in the case of a heel strike and a resulting soft feel upon impact, spacing between adjacent ones of the heel protrusions may be greater than spacing between adjacent ones of the midfoot protrusions. More space between protrusions enables greater “movement” or outward spread of the protrusions under compression without interference from neighboring protrusions (e.g., lower compressive stiffness). The relatively lower width-to-length ratio of the heel protrusions enables greater deformation regardless of an exact impact angle or location in the heel region of initial ground contact in a heel strike.

(79) When the sole structure rolls forward so that the midfoot protrusions come into contact with the horizontal ground plane, the transversely-elongated shape of a midfoot protrusion may cause it to compress down upon its front half, rolling over its peak, providing forward momentum as it straightens upon decompression as compressive force of the foot moves forward to the forefoot region. This may be referred to as longitudinal shear. Additionally, the transversely-elongated shape of the midfoot protrusions make them more resistant to transverse shear under transverse (medial-lateral) loading (e.g., when the sole structure is worn on the “outside” foot during a turn).

(80) In an example configuration, the midsole may be a foam body and the midfoot protrusions may be a ground contact surface of the foam body, such as when the weight of the foot is centered over the midfoot region so that the midfoot protrusions are in contact with the ground. In the same or another configuration, the heel protrusions may be a ground contact surface of the foam body, such as when the weight of the foot is centered over the heel region. Stated differently, the ground-facing surface of the midsole may be the ground-contact surface, the midsole thereby also serving the function of an outsole where the ground-facing surface is also the ground-contact surface. For example, no outsole element(s) may be secured to the ground-facing surface of the midsole at the midfoot protrusions and or at the heel protrusions in such configurations.

(81) In a configuration, a sole structure for an article of footwear may comprise a midsole having a ground-facing surface with a forefoot region, a midfoot region, and a heel region. The midsole may define downwardly-extending protrusions distributed over the midfoot region and the heel region, and the forefoot region may be relatively flat. The downwardly-extending protrusions may have convex outer surfaces and may transition in a forward direction from relatively round to relatively oblong and back to relatively round, each of the relatively oblong downwardly-extending protrusions having a width in a transverse direction of the midsole greater than a length in a longitudinal direction of the midsole. The relatively oblong downwardly-extending protrusions

may be taller than the relatively round downwardly-extending protrusions. The midsole may arc upward from the midfoot region to the heel region and upward from the midfoot region to the forefoot region.

(82) The above features and advantages and other features and advantages of the present teachings are readily apparent from the following detailed description of the modes for carrying out the present teachings when taken in connection with the accompanying drawings.

(83) Referring to the drawings, wherein like reference numbers refer to like components throughout the views, FIG. 1 shows a sole structure **10** for an article of footwear, such as the article of footwear **12** of FIGS. 7-9. FIG. 1 is a bottom view of the sole structure **10**. The sole structure **10** includes a midsole **14** and an outsole **16**, which, in the embodiment shown, is a single outsole element **16** secured to the midsole **14**. The sole structure **10** may include additional components, such as one or more sole layers overlaying the midsole **14** (shown in FIG. 7) and/or one or more bladders that retain gas. For example, a heel bladder may rest on the foot-facing surface **24** of the midsole **14**, or nest within the midsole **14** at a heel region **26** of the midsole **14**. FIG. 7 shows an upper **18** coupled to the sole structure **10**, and a foot **20** shown in phantom received in a foot-receiving cavity **22** defined by the upper **18** and the sole structure **10** and resting on a foot-facing surface **24** of the sole structure **10**. A strobil and/or an insole (not shown) may overlay the midsole and be disposed between the foot **20** and the midsole **14**. For purposes of discussion herein, however, the foot-facing surface **24** is a top surface of the midsole **14**, and the foot **20** is shown on the foot-facing surface **24** of the midsole **14**. The foot-facing surface **24** is shown as generally cupping the shape of the bottom of the foot **20**. Additionally or in the alternative, the foam of the midsole **14** could be shaped to provide additional geometry at the foot-facing surface **24**.

(84) As shown in FIG. 1, the sole structure **10** includes a heel region **26**, a midfoot region **28**, and a forefoot region **30**. The heel region **26** generally includes portions of the sole structure **10** corresponding with rear portions of a human foot, including the calcaneus bone, when the human foot of a size corresponding with the sole structure **10** is supported on the sole structure **10** as shown in FIG. 7. The forefoot region **30** of the sole structure **10** generally includes portions of the sole structure **10** corresponding with the toes and the joints connecting the metatarsals with the phalanges of the human foot (interchangeably referred to herein as the “metatarsal-phalangeal joints” or “MPJ” joints). The midfoot region **28** of the sole structure **10** is disposed between the heel region **26** and the forefoot region **30** and generally includes portions of the sole structure **10** corresponding with an arch area of the human foot, including the navicular joint.

(85) Also, various example features and aspects of the footwear **12** may be disclosed or explained herein with reference to a “longitudinal direction” and/or with respect to a “longitudinal length” of the footwear sole structure **10**. As shown in FIG. 1, the “longitudinal direction” is determined as the direction of a line LM extending from a rearmost heel location (RH in FIG. 1) to the forwardmost toe location (FT in FIG. 1) of the sole structure **10**. This line LM may also be referred to as the longitudinal midline. The “longitudinal length” L is the length dimension measured from the rearmost heel location RH to the forwardmost toe location FT. The rearmost heel location RH and the forwardmost toe location FT may be located by determining the rear heel and forward toe tangent points with respect to front and back parallel vertical planes VP when the sole structure **10** is oriented on a horizontal support surface such as a ground plane G shown in FIG. 7 in an unloaded condition (e.g., with no weight applied to the sole structure **10** other than the weight of the shoe components of the article of footwear **12**, such as the upper **18**). If the forwardmost and/or rearmost locations of a specific sole structure constitute a line segment (rather than a tangent point), then the forwardmost toe location and/or the rearmost heel location constitute the midpoint of the corresponding line segment. If the forwardmost and/or rearmost locations of a specific sole structure constitute two or more separated points or line segments, then the forwardmost toe location and/or the rearmost heel location constitute the midpoint of a line segment connecting the furthest spaced and separated points and/or furthest spaced and separated end points of the line

segments (irrespective of whether the midpoint itself lies on the sole structure **10**). If the forwardmost and/or rearwardmost locations constitute one or more areas, then the forwardmost toe location and/or the rearwardmost heel location constitute the geographic center of the area or combined areas (irrespective of whether the geographic center itself lies on the sole structure **10**).

(86) Once the longitudinal direction of the sole structure **10** has been determined with the sole structure **10** oriented on the horizontal ground plane **G**, planes may be oriented perpendicular to this longitudinal direction (e.g., planes running into and out of the page of FIG. **1**). The locations of these perpendicular planes may be specified based on their positions along the longitudinal length **L** where the perpendicular plane intersects the longitudinal direction between the rearmost heel location **RH** and the forwardmost toe location **FT**. In this illustrated example of FIG. **1**, the rearmost heel location **RH** is considered as the origin for measurements (or the “0 **L** position”) and the forwardmost toe location **FT** is considered the end of the longitudinal length of this component (or the “1.0 **L** position”). Plane position may be specified based on the plane's location along the longitudinal length **L** (between 0 **L** and 1.0 **L**), measured forward from the rearmost heel **RH** location in this example. FIG. **1** further shows locations of various planes perpendicular to the longitudinal direction (and oriented in the transverse direction) and located along the longitudinal length **L** at positions 0.3 **L** and 0.6 **L** (measured in a forward direction from the rearmost heel location **RH**). These planes may extend into and out of the page of the paper from the view shown in FIG. **1**, and similar perpendicular planes may be oriented at any other desired positions along the longitudinal length **L**. While these planes may be parallel to the parallel vertical planes **VP** used to determine the rearmost heel **RH** and forwardmost toe **FT** locations, this is not a requirement. Rather, the orientations of the perpendicular planes along the longitudinal length **L** will depend on the orientation of the longitudinal direction, which may or may not be parallel to the horizontal ground plane **G** in the arrangement/orientation shown in FIG. **1**.

(87) As shown in FIG. **1**, the heel region **26** of the sole structure **10** is defined herein as being between perpendicular planes located at 0 **L** and 0.3 **L** of the sole structure **10**, the midfoot region **28** of the sole structure **10** is defined herein as being between perpendicular planes located at 0.3 **L** and 0.6 **L**, and the forefoot region **30** of the sole structure **10** is defined herein as being between perpendicular planes located at 0.6 **L** and 1.0 **L**.

(88) The sole structure **10** has a medial side **32** (also shown in FIG. **4**) and a lateral side **34** (also shown in FIG. **3**) both of which extend from the heel region **26** to the forefoot region **30** and are generally on opposite sides of the longitudinal midline **LM** of the sole structure **10**. The medial side **32**, the lateral side **34**, and a rear **36** of the sole structure **10** described herein correspond with and may also be used to indicate the medial side, the lateral side, and the rear of individual components of the sole structure **10**, such as of the midsole **14**.

(89) The midsole **14** has a ground-facing surface **40**, portions of which fall within the forefoot region **30**, the midfoot region **28**, and the heel region **26**, as shown in FIGS. **1** and **2**. In FIG. **2**, the outsole element **16** is removed. The midsole **14** has a slight lip **42** disposed in the ground-facing surface **40** on the forefoot region **30**. When the outsole element **16** is secured to the ground-facing surface **40** in the forefoot region **30**, such as with adhesive and/or thermal bonding, a rear extent of the outsole element **16** abuts the lip **42**. The forefoot region **30** has a forward half (e.g., from the vertical plane 0.8 **L** to the vertical plane **VP** at the forwardmost toe location **FT**) and a rear half (e.g., from the vertical plane at 0.6 **L** to the vertical plane at 0.8 **L**). In the embodiment shown, the outsole element **16** covers much of the ground-facing surface **40** in the forward half of the forefoot region **30** and a majority of the rear half is uncovered. In other configurations an outsole element may cover more or less of the forefoot region **30** and/or more than one outsole element may be secured to the ground-facing surface **40**.

(90) The midsole **14** defines downwardly-extending protrusions **44** at the ground-facing surface **40** distributed over the midfoot region **28** and the heel region **26** and absent from at least the forward half (e.g., from a vertical plane at 0.8 **L** to the vertical plane **VP** at the forwardmost toe location **FT**

(at 1.0 L)) of the ground-facing surface **40** of the forefoot region **30**. For example, the outsole element **16** covers the ground-facing surface **40** of the midsole **14** only in a portion of the forefoot region **30**, and forwardmost protrusions **44C** of the downwardly-extending protrusions **44** are rearward of a widest portion **47** of the midsole **14** in the forefoot region **30**. A majority of the outsole element **16** is forward of the widest portion **47** of the midsole **14** in the forefoot region **30**. Providing the outsole element **16** in the forefoot region **30** may enable increased durability and traction needed for toe-off (shown in FIG. **9**). In some embodiments, the material of the midsole **14** alone may provide sufficient durability and traction not only in the midfoot region **28** and heel region **26**, but also in the forefoot region **30** such that no outsole element is included in the sole structure **10**.

(91) Each of the downwardly-extending protrusions **44** has a convex outer surface **45** as best shown in FIG. **2**. Only some of the convex outer surfaces **45** are labelled in FIG. **2** for clarity in the drawings. The downwardly-extending protrusions **44** include midfoot protrusions **44B** in the midfoot region **28** having widths in a transverse direction of the midsole **14** greater than lengths in a longitudinal direction of the midsole **14**.

(92) The downwardly-extending protrusions **44** also include heel protrusions **44A** in the heel region **26** having width-to-length ratios less than width-to-length ratios of the midfoot protrusions **44B**. For example, the midfoot protrusions **44B** may be relatively oblong and the heel protrusions **44A** may be relatively round. In determining a width of a protrusion **44**, a measurement is taken perpendicular to the longitudinal midline LM and measuring between the points spaced furthest apart from one another in the transverse direction and falling on an outer perimeter OP of the protrusion **44** (e.g., where an outer perimeter OP falls along an outline of a change in curvature where the protrusion **44** begins extending downward from an overlaying base portion **14B** of the midsole **14**). Some of the outer perimeters OP are labelled in FIG. **1**, and each appears as a closed curve surrounding a protrusion **44**. The base portion **14B** can be seen as the spaces between the protrusions **44** at the ground-facing surface **40** in FIG. **1**. In determining a length of a protrusion **44**, a measurement is taken parallel to the longitudinal midline LM and measuring between the points spaced furthest from one another in the longitudinal direction and falling on the outer perimeter OP of the protrusion **44** at the ground-facing surface **40**. In referencing a protrusion **44** as being generally oblong or generally round, the reference is with respect to the shape of the outer perimeter OP of the protrusion.

(93) It is noted that not all of the downwardly-extending protrusions **44** in the midfoot region **28** need have widths greater than lengths and/or need be generally oblong in the transverse direction in order to fall within the scope of the disclosure and fulfill the advantages of the sole structure **10** as discussed herein. Additionally, not all of the downwardly-extending protrusions **44A** in the heel region **26** need have width-to-length ratios less than the width-to-length ratios of the midfoot protrusions **44B** in order to fall within the scope of the disclosure and fulfill the advantages of the sole structure **10** as discussed herein. However, as can be seen in FIG. **1**, more of the downwardly-extending protrusions **44A** in the heel region **26** are generally round and have width-to-length ratios less than the width-to-length ratios of the midfoot protrusions **44B**, more of which are generally oblong.

(94) One example generally oblong midfoot protrusion **44B1** is indicated having a width W1 and a length L1 in FIG. **1**. One example generally round heel protrusion **44A1** is indicated having a width W2 and a length L2 in FIG. **1**. It can be seen that the ratio of the width W2 to the length L2 of the heel protrusion **44A1** is less than the ratio of the width W1 to the length L1 of the midfoot protrusion **44B1**. The heel protrusions **44A** include but are not limited to those protrusions that are the rearmost protrusions **44A1** and **44A2** not truncated at an edge of the midsole **14** as discussed herein. The midfoot protrusions **44B** include but are not limited to those protrusions **44B1**, **44B2**, and **44B3** not truncated at an edge of the midsole **14**. Additionally, the downwardly-extending protrusions **44** include forwardmost protrusions **44C** that may fall within the rear half of the

forefoot region **30** and/or may be at a forward portion of the midfoot region **28**. Especially viewing the protrusions **44** from the rearmost protrusions not truncated at an edge of the midsole **14** (e.g., protrusions **44A1** and **44A2**) to the forwardmost protrusions **44C** not truncated at an edge of the midsole **14**, the downwardly-extending protrusions **44** that fall along the longitudinal midline LM generally transition in a forward direction from relatively round to relatively oblong and back to relatively round while increasing in height and then decreasing in height. Shorter protrusions are more neutral in terms of the exact location in the heel region **26** that is the first point of contact to lessen the potential of side-to-side movement and an off-balance feeling upon heel strike.

(95) The midsole **14** may be a foam body such as a foamed polymeric material. In some embodiments, the midsole **14** may be at least partially a polyurethane (PU) foam, a polyurethane ethylene-vinyl acetate (EVA) foam, and may include heat-expanded and molded EVA foam pellets. In some examples, the foam material may comprise a blend of EVA material or materials, for example. The midsole **14** may comprise Pebax® thermoplastic elastomer foam and may be sold under the tradename ZoomX by Nike, Inc. The outsole element **16** may include a rubber material than may be a natural rubber, or a synthetic rubber, or a combination of both. Examples of types of rubbers include butadiene rubber, styrene-butadiene (SBR) rubber, butyl rubber, isoprene rubber, urethane rubber, nitrile rubber, neoprene rubber, ethylene propylene diene monomer (EPDM) rubber, ethylene-propylene rubber, urethane rubber, polynorbornene rubber, methyl methacrylate butadiene styrene (MBS) rubber, styrene ethylene butylene (SEBS) rubber, silicone rubber, and mixtures thereof. The rubber compound may be a virgin material, a regrind material, and mixtures thereof.

(96) As the outsole element **16** is disposed only in the forefoot region **30** in the embodiment shown, the convex outer surfaces **45** of the midfoot protrusions **44B**, the heel protrusions **44A**, and the forwardmost protrusions **44C** serve as a ground contact surface of the sole structure **10**. For example, some or all of the convex outer surfaces **45** of the heel protrusions **44A** as well as the truncated peripheral protrusions **44D** in the heel region **26** are a ground contact surface of the sole structure **10**, such as during a heel strike and/or when the weight of the foot **20** is centered over the heel region **26** and the heel protrusions **44A** are in contact with the horizontal ground plane G, as shown in FIG. 7. Similarly, some or all of the convex outer surfaces **45** of the midfoot protrusions **44B** and the truncated peripheral protrusions **44D** in the midfoot region **28** are the ground contact surface of the sole structure **10**, such as when the weight of the foot **20** is centered over the midfoot region **28** so that the midfoot protrusions **44B** are in contact with the ground plane G as shown in FIG. 8. Stated differently, the midsole **14** also serves the function of an outsole in at least the heel region **26** and the midfoot region **28** as the convex outer surfaces **45** of the midfoot protrusions **44B** and the heel protrusions **44A** as well as the truncated peripheral protrusions **44D** are exposed (e.g., uncovered), with no outsole elements secured thereto in the embodiment shown. Alternatively, in some configurations, one or more outsole elements may be secured to the ground-facing surface **40** and/or to the convex outer surfaces **45** of some or all of the downwardly-extending protrusions **44** in the midfoot region **28** and/or the heel region **26**.

(97) The midsole **14** is shown as a one-piece foam body. For example, each of the downwardly-extending protrusions **44** and the base portion **14B** from which they extend are a unitary, one-piece (e.g., single) component. For example, a foam material may be injection molded, compression molded, or otherwise manufactured as the midsole **14** that is as a one-piece component. In an alternative embodiment, the midsole **14** with the shape shown and described herein including the protrusions of the shapes and sizes shown and described herein could be a fluid-filled bladder that defines an interior cavity and is configured to retain a fluid in the interior cavity. For example, polymeric sheets may be secured to one another at a peripheral flange to enclose the interior cavity and retain a fluid in the interior cavity, or a single polymeric sheet may be folded on itself to define a sealed peripheral flange, or polymeric material in a preform that is not a sheet may be blow-molded to define the bladder. As used herein, a “fluid” filling the interior cavity may be a gas, such

as air, nitrogen, another gas, or a combination thereof. The polymeric material when formed and inflated may define the protrusions **44**.

(98) As indicated in FIG. **1** and best shown in the closeup view of FIG. **6**, the forwardmost protrusions **44C** each have a front half **46** and a rear half **48**, and the convex outer surface **45** (e.g., the portion of the ground-facing surface **40** at the protrusion **44C**) is steeper at the rear half **48** than at the front half **46**. The front half **46**, being less steep, more gradually extends forward in the forefoot region **30** of the ground-facing surface. The ground-facing surface **40** in the forefoot region **30** has an absence of the downwardly-extending protrusions forward of the forwardmost protrusions **44C**, and is relatively flat in comparison to the ground-facing surface **40** in the midfoot region **28** and the heel region **26**.

(99) As best shown in FIG. **2**, the downwardly-extending protrusions **44** taper in width and length from the base portion **14B** to peaks **50**. Only some of the peaks **50** are labelled in FIGS. **1** and **2**. When a protrusion **44** contacts the horizontal ground plane G, the peak **50** comes into contact with the horizontal ground plane G first, and the protrusion **44** is compressed so that more surface area of the protrusion (e.g., more of the convex outer surface **45**) comes into contact with the horizontal ground plane G as loading progresses. Pressure is concentrated at the peak **50** by the initial load due to the relatively small surface area at the peak **50**, and lessens as the surface area broadens with compression of the protrusion **44**. The distribution of the load over the increasing surface area and the resultant reduction in pressure causes the rate of deflection of the protrusion **44** to decrease with time. Stated differently, due to the shape of the protrusion **44** broadening from the relatively narrow peak **50**, the protrusion **44** initially compresses more rapidly and gradually the rate of deflection decreases. This contrasts with a relatively low rate of deflection that would be experienced with a sole structure having a flatter area initially coming into contact with the horizontal ground plane G.

(100) As shown in FIGS. **2-6**, the relatively oblong downwardly-extending protrusions **44B** of the midfoot region **28** are taller than the relatively round downwardly-extending protrusions **44A** of the heel region **26**. That is, as best shown in FIG. **5**, the height H1 of the midfoot protrusion **44B** from a plane extending through its outer perimeter OP (outer perimeters indicated in FIG. **1**) is greater than the height H2 of the heel protrusion **44A** from a plane extending through its outer perimeter OP. The taller midfoot protrusions **44B** are able to “lean” during a forward stride, as discussed with respect to FIG. **8**.

(101) As shown in FIGS. **3-5**, the midsole **14** arcs upward from the midfoot region **28** to the heel region **26**, and upward from the midfoot region **28** to the forefoot region **30**. Referring to FIG. **5**, a surface S tangent to the peaks **50** of the protrusions **44** and tangent to the lower surface of the outsole element **16** remains generally level at any given transverse cross section perpendicular to the longitudinal midline LM, and arcs from the rearmost extent of the midsole **14** (e.g., at the rearmost heel location RH) to a forward extent of the midsole **14** (near the forwardmost toe location FT), establishing an arced profile of the midsole with the midfoot region **28** being lower than the heel region **26** and forefoot region **30** and therefore in contact with the ground plane G when the sole structure **10** is unloaded. As discussed with respect to FIGS. **7-9**, this configuration helps to maintain a more constant forward angular momentum (indicated by arrows AM) during a forward stride than a sole structure without such an arced curvature. Such an embodiment enables a “rocker” functionality of the midsole **14**. With such a full-length, convex camber, only a relatively small area of the ground-facing surface **40** is in contact with the ground plane G at any time during the stride. This helps to avoid a “slapping” phenomena and associated foot fatigue that may occur with sole structures configured so that a large area of the midfoot region comes into contact with the ground plane abruptly upon transition from the heel region to the midfoot region, for example.

(102) FIGS. **2, 3** and **14** also best show that the midsole **14** includes a rear side wall **52** that flares outward from an upper extent **52A** to a lower extent **52B** of the rear side wall **52** at the rear **36** of the heel region **26**. This flare extends the lower extent **52B** more rearward than if the rear side wall **52** were more perpendicular to the horizontal ground plane G. This may help to steer the midsole

14 into the forward rocking motion early in a heel strike of a forward stride.

(103) Similarly, as best shown in FIGS. **12-15**, the midsole **14** includes a lateral side wall **58** and a medial side wall **56**, each of which has an upper extent and a lower extent and each of which flares outward from the upper extent to the lower extent in both the forefoot region **30** (see FIGS. **13** and **15**) and the heel region **26** (see FIGS. **14** and **15**). Specifically, as shown in FIG. **15**, the medial side wall **56** flares outward from an upper extent **56A** to a lower extent **56B**. The lateral side wall **58** flares outward from an upper extent **58A** to a lower extent **58B**. As is evident in FIGS. **1** and **15**, the midsole **14** is widest in the forefoot region **30** generally just forward of the forwardmost protrusions **44C** and approximately where the rear edge **16A** of the outsole element **16** is secured to the midsole **14**. The heel region **26** is also relatively wide generally just forward of the rearmost (non-truncated) heel protrusion **44A1** labelled in FIG. **1**. The relatively wide and flat forefoot region **30** provides a stable platform for toe-off. The height of the midsole **14** in the forefoot region **30** is sufficient to provide adequate cushioning, while the relative flatness of the midsole **14** in the forefoot region **30** (e.g., relative to the midfoot region **28** and the heel region **26** due to the shorter downwardly-extending protrusions **44** and the absence of downwardly-extending protrusions in the forward portion of the forefoot region **30**) makes this region relatively stiff in comparison to the other regions **28**, **30** to provide support for an efficient toe-off.

(104) As shown in FIGS. **1-4**, the downwardly-extending protrusions **44** include truncated peripheral protrusions **44D** in the midfoot region **28** and the heel region **26** that define a lateral side edge **34A** and a medial side edge **32A** of the ground-facing surface **40** of the midsole **14**. Similar truncated peripheral protrusions **44D** define a rear edge **36A**. The peripheral protrusions **44D** are truncated in that they each have only about half of a convex outer surface **45** in comparison to the midfoot protrusions **44B** and heel protrusions **44A**, as peaks **50** of the truncated peripheral protrusions **44D** lie along the lateral side edge **34A**, the medial side edge **32A**, and the rear edge **36A**. Providing peaks **50** along the edges **32A** and **34A** broadens the width between the two furthest peaks **50** in contact with the ground plane **G** during the forward stride, increasing medial-lateral stability. Providing peaks **50** along the rear edge **36A** and edges **32A**, **34A** helps to limit the extent of medial-lateral tilt that could occur regardless of whether initial impact is on a non-truncated, fully convex protrusion **44A** or **44B**.

(105) To promote resilient deformation in the case of a heel strike and resulting soft feel upon impact, spacing between adjacent ones of the heel protrusions **44A** may be greater than spacing between adjacent ones of the midfoot protrusions **44B**. For example, FIG. **1** shows, on average, more of the ground-facing surface **40** disposed between outer peripheries **OP** of the heel protrusions **44A** (e.g., spacing **S1**) than the midfoot protrusions **44B** (e.g., spacing **S2**). More space between heel protrusions **44A** enables greater “movement” or outward spread of the heel protrusions **44A** under compression by compressive forces **F** without interference from neighboring protrusions (e.g., lower compressive stiffness). The lower width-to-length ratio of the heel protrusions **44A** enables greater deformation regardless of an exact impact angle or location in the heel region **26** of initial ground contact in a heel strike, shown in FIG. **7**.

(106) FIGS. **7-9** show three instants in time during three different phases of the forward roll of the sole structure **10**, with the curved arrows **AM** indicating the angular momentum of the sole structure **10**. As shown in FIG. **7**, due to the arced profile of the sole structure **10** discussed herein, only the heel region **26** is in contact with the ground plane **G** during the initial heel strike.

(107) As forward movement progresses, the sole structure **10** rolls forward so that the midfoot protrusions **44B** are in contact with the horizontal ground plane **G**. Because of the arced profile of the sole structure **10**, the heel region **26** and the forefoot region **30** are not in contact with the ground plane **G**. The relatively tall height of the transversely-elongated midfoot protrusions **44B** may cause them to compress down upon their front sides, rolling over their peaks **50** as illustrated in FIG. **8** (e.g., leaning or bending over onto their front sides), each providing forward momentum as it straightens upon decompression as compressive force **F** of the foot **20** moves forward to the

forefoot region **30**. The leaning or bending movement may be referred to as longitudinal shear. The relatively tall height of the midfoot protrusions **44B** encourages and enables this shearing in the longitudinal direction. However, the transversely-elongated shape of the midfoot protrusions **44B** make them more resistant to transverse shear under transverse loading which may occur when turning directions while loading the sole structure **10** (e.g., when the sole structure **10** is worn on the “outside” foot **20** during a turn).

(108) FIG. **9** depicts the sole structure **10** during a further forward roll as the ground-facing surface **40** of the forefoot region **30** becomes the ground contact surface during the forward stride. Because of the arced profile of the sole structure **10**, the heel region **26** and the midfoot region **28** are no longer in contact with the ground **G**. The absence of downwardly-extending protrusions **44** in at least the forward half of the forefoot region **30** creates a relatively stiff forefoot as more surface area in the transverse direction is in contact with the ground plane **G** at any point along the length of the sole structure **10** in the forward roll. Because the entire width of the midsole **14** in the forefoot region **30** is relatively flat in comparison to the protrusions **44**, more surface area of the forefoot region **30** may be in contact with the horizontal ground plane **G** in comparison to just portions of the convex outer surfaces **45** of protrusions **44**, so that the forefoot region **30** contributes to a supportive platform for ease of toe-off. To smoothly transition to the more stable platform of the relatively flat portion of the forefoot region **30**, the shape of the protrusions **44** goes from relatively elliptical back to relatively round (e.g., at protrusions **44C**) and the height of the forwardmost protrusions **44C** decreases gradually to zero (e.g., with the forward halves **46** of the forwardmost protrusions **44C** being less steep than the rear halves **48**).

(109) Accordingly, the compressibility of the protrusions **44** mitigates impact, while the arced profile of the sole structure **10** promotes an efficient transition from heel strike to toe-off. Typically, it is difficult to achieve both of these goals, as increasing compressibility often decreases the efficiency of forward motion (e.g., the wearer must work relatively harder to maintain forward momentum in a highly compressible, cushioned midsole without the arced profile from the heel region **26** to the forefoot region **30**). The arced profile helps maintain a more constant angular momentum from the heel strike position of FIG. **7** to the midfoot position of FIG. **8** in comparison to the abrupt increase in angular momentum that may occur with a more typical profile of a sole structure (e.g., a sole structure without an arc upward extending from the midfoot region **28** to the rear of the heel region **26**).

(110) FIG. **10** is a cross-sectional illustration of the sole structure **10** taken at lines **10-10** in FIG. **15**. A relatively short protrusion **44C** at the forward portion of the midfoot region **28** is shown, with the cross-section taken forward of the peak **50** so that the full height is not apparent. FIG. **11** is a cross-sectional illustration of the sole structure **10** taken at lines **11-11** in FIG. **15**. The flaring side wall **56** and the truncated peripheral protrusions **44D** are apparent, as well as the relatively tall height of the protrusion **44B3** in comparison to the more forward protrusion **44C** of FIG. **10**.

(111) FIG. **12** is a cross-sectional illustration of the sole structure taken at lines **12-12** in FIG. **15**. The relatively rounder and shorter protrusion **44A1** is shown, and the cross-section is taken rearward of the peak of protrusion **44A2** and illustrates the partially truncated peripheral protrusion **44D**. As the rearmost truncated peripheral protrusions **44D** are disposed higher than the protrusion **44A1** due to the arced profile apparent in FIG. **3**, these are not visible in the cross-section taken.

(112) FIG. **13** is a front view of the sole structure **10**, showing the lateral side wall **58** flaring outward and the medial side wall **56** flaring outward. The outsole element **16** is shown with a rear extent generally at the widest portions **47** of the midsole **14**. The medial side wall **56** and lateral side wall **58** flaring outward from an upper extent to a lower extent are apparent in the forefoot region **30** shown in FIG. **13**.

(113) FIG. **14** is a rear view of the sole structure **10**, showing the truncated peripheral protrusions **44D** providing a broad platform in the transverse direction to avoid medial-lateral motion. FIG. **15** is a top view of the sole structure **10** indicating the flaring medial and lateral side walls **56**, **58** and

rear side wall **52** in the heel region **26**.

(114) FIG. **16** shows another embodiment of a sole structure **110** that may be used in the article of footwear **12** of FIG. **7** in place of sole structure **10**. The sole structure **110** has many of the same features as described with respect to the sole structure **10**, and these features are indicated with like reference numbers. Like sole structure **10**, the sole structure **110** includes a midsole **114** that has a plurality of downwardly-extending protrusions **44** (see FIGS. **17-19** and **24**, for example) that mitigate impact, and has an arced profile to promote efficient transition from heel strike to toe-off. As further discussed herein, the sole structure **110** also includes an outsole **116** covering at least a portion of the ground-facing surface **40** of the midsole **114**. In the embodiment shown, the outsole **116** has a first outsole element **116A**, a second outsole element **116B**, and a third outsole element **116C**, also referred to herein as a first outsole portion, a second outsole portion, and a third outsole portion, respectively.

(115) The first outsole element **116A** is secured to and covers at least part of the ground-facing surface **40** of the midsole **114** in the forefoot region **30**. The first outsole element **116A** is also referred to as a forefoot outsole element **116A**. The second outsole element **116B** is secured to and covers at least part of the ground-facing surface **40** of the midsole **114** in the heel region **26**. The second outsole element **116B** is also referred to as a heel outsole element **116B**. The third outsole element **116C** is secured to and covers at least part of the ground-facing surface **40** of the midsole **114** in the midfoot region **28**. The third outsole element **116C** is also referred to as a midfoot outsole element **116C**. As best shown in FIGS. **28-31**, each of the forefoot outsole element **116A**, the midfoot outsole element **116C**, and the heel outsole element **116B** extend from the medial side wall **56** to the lateral side wall **58** of the midsole **114**.

(116) As shown in FIG. **16**, the first outsole element **116A** and the third outsole element **116C** are separated by a first gap **151** extending from a medial edge **132A** to a lateral edge **134A** of the midsole **114**. The medial edge **132A** is also referred to as a medial side edge of the ground-facing surface **40**, and the lateral edge **134A** is also referred to as a lateral side edge of the ground-facing surface **40**. FIGS. **20**, **21**, and **22** also depict the first gap **151**. Stated differently, a rear edge **153** of the first outsole element **116A** (e.g., the forefoot outsole element) is spaced apart from a forward edge **155** of the third outsole element **116C** (e.g., the midfoot outsole element) defining the first gap **151** between the rear edge **153** and the forward edge **155**.

(117) The rear edge **153** of the first outsole element **116A** has an irregular shape. As used herein, an irregular shape is a nonlinear shape. The forward edge **155** of the third outsole element **116C** has a complementary irregular shape that tracks the irregular shape of the rear edge **153** of the first outsole element **116A**. As used herein, an edge has a complementary irregular shape that tracks an irregular shape of another edge when the edges can be spaced apart from one another by a gap of a substantially constant width. For example, a substantially constant width of a gap may be a gap with a width that varies by not more than **20** percent along a length of the gap. As shown in FIGS. **16**, **20**, **21**, and **22**, the first gap **151** has a substantially constant width.

(118) Similarly, the second outsole element **116B** and the third outsole element **116C** are separated by a second gap **157** extending from the medial edge **132A** to the lateral edge **134A** of the midsole **114**. Stated differently, a rear edge **159** of the third outsole element **116C** (e.g., the midfoot outsole element) is spaced apart from a forward edge **161** of the second outsole element **116B** (e.g., the heel outsole element) defining the second gap **157** between the rear edge **159** and the forward edge **161**.

(119) The forward edge **161** of the second outsole element **116B** has an irregular shape, and the rear edge **159** of the third outsole element **116C** has a complementary irregular shape that tracks the irregular shape of the forward edge **161** of the heel outsole element. Accordingly, the second gap **157** is of a substantially constant width.

(120) As best shown in FIGS. **18**, **19** and **24**, the midsole **114** includes a first ridge **170** and a second ridge **172**. The first ridge **170** and the second ridge **172** protrude downward and form part of

the ground-facing surface **40**. The first ridge **170** and the second ridge **172** extending downward from the contoured downwardly-extending protrusions **44** by an amount **H3**, **H4**, respectively that may vary along each of the ridges **170**, **172**. Stated differently, the ridges **170**, **172** follow the contours of the downwardly-extending protrusions **44** in the vertical direction while also following the contours of the gaps **151**, **157** respectively, in the horizontal direction.

(121) As shown in FIGS. **16**, **17**, and **24**, the first ridge **170** extends into and at least partially fills the first gap **151**. Similarly, the second ridge **172** extends into and at least partially fills the second gap **157**.

(122) The configuration of the outsole **116** having two or more discrete outsole elements separated from one another (such as any two of or all of the outsole elements **116A**, **116B**, and **116C**) constrains the midsole **114** less than would an embodiment in which a one-piece outsole extends across most or all of the ground-facing surface **40** of the midsole **114**. The midsole **114** as shown in a one-piece foam body that may be any of the materials described with respect to midsole **14**. The outsole **116** may be any of the materials described with respect to outsole **16**. The midsole **114** may thus be of a material that is more compressible and flexible than the outsole **116**. The gaps **151**, **157** allow greater movement of the midsole **114** during compression and dorsiflexion, for example, than if the outsole **116** extended without gaps along the ground-facing surface **40** from the rear extent of the heel region **26** (e.g., the rearmost heel location **RH**) to the front extent of the forefoot region **30** (e.g., the foremost toe location **FT**). By providing edges of adjacent outsole elements that are complementary and track one another, the competing goals of covering much of the ground-facing surface **40** of the midsole **114** with the outsole **116** to increase durability and allowing flexibility and deformation of the protrusions **44** of the midsole **114** without excessive constraint by the outsole **116** may both be achieved.

(123) Referring to FIG. **24**, a height **H5** of the midsole **114** is greatest at the midfoot region **28**. The height of the midsole **114** varies given the undulating protrusions **44**, but is consistently greater at the peaks **50** of the protrusions **44** in the midfoot region **28** than at the peaks **50** of the protrusions **44** in the heel region **26**. The height of the midsole **114** at any location is measured from the foot-facing surface to the ground-facing surface **40**. The height of the midsole **114** is also greater at the peaks **50** of the protrusions **44** in the midfoot region **28** than in the forefoot region **30**. The outsole **116** is configured so that the height **H6** of the sole structure **110** is also greatest at the midfoot region **28**. Stated differently, the height of the sole structure **110**, including the height of both the midsole **114** and the outsole **116**, is greatest at the midfoot region **28**. Accordingly, like the midsole **114**, the sole structure **110** (including both the midsole **114** and the outsole **116**) also has an arced profile.

(124) Such an embodiment enables a “rocker” functionality of the midsole **114**. For example, the ground-facing surface **40** of the midsole **114** curves upwardly from the midfoot region **28** to a forward extent of the midsole **114** and from the midfoot region **28** to a rear extent of the midsole **114**, establishing an arced profile of the midsole **114**. As discussed with respect to midsole **14**, with such a full-length, convex camber, only a relatively small area of the ground-facing surface of the outsole **116** is in contact with a horizontal ground plane at any time during the stride, and the rate of transitioning forward on the sole structure **110** is relatively constant in comparison to sole structures configured so that a large portion of the midfoot region **28** comes into contact with the ground very abruptly upon transition from a heel region **26** to a midfoot region **28**, for example. The full-length convex camber helps to avoid a “slapping” phenomena and associated foot fatigue.

(125) With reference to FIGS. **19** and **24**, in contrast to the midfoot region **28** and heel region **26** having the downwardly-extending protrusions **44**, the ground-facing surface **40** of the forefoot region **30** is relatively flat. To provide a stable platform for toe-off from the forefoot region **30**, downwardly-extending protrusions **44** may be absent from at least a forward half of the forefoot region **30**.

(126) Referring again to FIGS. **19** and **24**, the downwardly-extending protrusions **44** are distributed

over the midfoot region **28** and the heel region **26**. Like midsole **14**, each of the downwardly-extending protrusions **44** has a convex outer surface **45** and a peak **50**. Only some of these are indicated with reference numbers in FIGS. **19** and **24**. In contrast to the midsole **14**, however, the downwardly-extending protrusions **44** in the midfoot region **28** of the midsole **114** have width-to-length ratios similar to those of the downwardly-extending protrusions in the forefoot region **30** and the heel region **26**. In other words, like those in the forefoot region **30** and heel region **26**, the protrusions **44** in the midfoot region **28** (e.g., midfoot protrusions **44B**) are relatively round rather than elongated. The protrusions **44** in the midfoot region **28** have widths in a transverse direction of the midsole **114** that are approximately the same as or may be equal to lengths in a longitudinal direction of the midsole **114**, and protrusions **44** in the heel region **26** (e.g., heel protrusions **44A**) have width-to-length ratios that are relatively the same as the width-to-length ratios of the protrusions **44** in the midfoot region **28**.

(127) Referring to FIG. **19**, like midsole **14**, the downwardly-extending protrusions **44** in the forefoot region **30** include forwardmost protrusions **44C** each having a front half **46** and a rear half **48**, and the convex outer surface **45** is steeper at the rear half **48** than at the front half **46**, enabling the forward roll of the foot to smoothly transition from the undulating midfoot region **28** to the relatively flat forefoot region **30**.

(128) As best shown in FIG. **24**, the outsole **116** is thicker where it aligns with and is secured to the peaks **50** of the downwardly-extending protrusions **44** than at the ground-facing surface **40** of the midsole **114** between adjacent peaks **50**. For example, the outsole **116** at areas A and B (at peaks **50**) is thicker than the outsole at areas C and D (between adjacent peaks **50**). The material of the outsole **116** may be relatively durable in comparison to the material of the midsole **114**, and the thickness at the peaks **50** aids in durability of the sole structure **110**, slowing wear. The material of the midsole **114** may be relatively more compressible than the outsole **116**, and the relative thinness of the outsole **116** between the peaks **50** may therefore enable greater movement and deformation of the downwardly-extending protrusions **44** during compression in comparison to a thicker outsole between the peaks **50**. Stated differently, an outsole **116** with relatively thin areas between the peaks **50** constrains the midsole **114** less than if the material between the peaks **50** were thicker, allowing greater resilient deformation and related cushioning. As is evident in FIGS. **21** and **22**, the outsole **116** is preformed with these relatively thick and thin areas, and with an undulating inner or top surface **74** (e.g., foot-facing surface **74**) having recesses **75** that align with and cup the protrusions **44** when the outsole **116** is assembled to the ground-facing surface **40** of the midsole **114**. Only some of the recesses **75** are labelled in FIGS. **21-22**. Because the outsole **116** is preformed with this three-dimensional, undulating shape rather than being a flat piece molded onto the midsole **114** to obtain the undulating shape, there may be less tendency for the protrusions **44** to be compressed or otherwise deformed by the outsole **116** during attachment of the outsole **116** to the midsole **114**, allowing the midsole **114** to thus function in accordance with its intended shape during use.

(129) Like midsole **14**, the midsole **114** includes a lateral side wall **58** and a medial side wall **56**, each of which has an upper extent and a lower extent and each of which flares outward from the upper extent to the lower extent in both the forefoot region **30** (see, e.g., FIGS. **25**, **26**, and **28**) and the heel region **26** (see, e.g., FIGS. **25**, **27**, and **31**). Specifically, the medial side wall **56** flares outward from an upper extent **56A** to a lower extent **56B**, and the lateral side wall **58** flares outward from an upper extent **58A** to a lower extent **58B**. Accordingly, the forefoot region **30** is relatively flat and wide to provide stability. The height of the midsole **114** in the forefoot region **30** should be sufficient to provide adequate cushioning, while the relative flatness of the foam (absence of the downwardly-extending protrusions **44**) makes this region relatively stiff in comparison to other regions to provide support for an efficient toe-off. Similarly, the heel region **26** flares outward to be sufficiently wide at the lower extents **56B**, **58B** of the side walls **56**, **58** to provide stability.

(130) Referring again to FIGS. **25-27**, the medial side wall **56** has a lower medial side edge **56C** at

the lower extent **56B** extending from the forefoot region **30** to the heel region **26**. The lateral side wall **58** has a lower lateral side edge **58C** at the lower extent **58B** extending from the forefoot region **30** to the heel region **26**. The outsole **116** (including outsole elements **116A**, **116B**, and **116C**) extends to and underlies the medial side edge **56C** and the lateral side edge **58C**, terminating at the medial side edge **56C** and the lateral side edge **58C** without extending onto the medial side wall **56** and the lateral side wall **58**. Stated differently, the outsole **116** does not wrap upward along the side walls **56**, **58**. By not extending onto the medial side wall **56** and the lateral side wall **58**, the midsole **114** is less constrained by the outsole **116** at the side walls **56**, **58** and may resiliently deform to a greater degree, potentially providing greater cushioning.

(131) Like midsole **14**, for stability in the midfoot region **28** given the convex outer surfaces **45** of the downwardly-extending protrusions **44**, the downwardly-extending protrusions **44** include truncated peripheral protrusions **44D** in the midfoot region **28** and the heel region **26** that define a lateral side edge **34A** and a medial side edge **32A** of the ground-facing surface **40** of the midsole **114**, as shown in FIG. **31**. The peripheral protrusions **44D** are truncated in that they each have only about half of a convex outer surface **45** in comparison to the midfoot protrusions **44B** and heel protrusions **44A** that are not truncated, as peaks **50** of the truncated peripheral protrusions **44D** lie along the lateral side edge **34A** and the medial side edge **32A**. Providing peaks **50** along the edges **32A** and **34A** broadens the width between the two furthest peaks **50** in contact with the ground plane **G** during the forward stride, increasing medial-lateral stability. Providing peaks **50** along the edges **32A**, **34A** helps to limit the extent of medial-lateral tilt that could occur regardless of whether initial impact is on a non-truncated, fully convex protrusion **44**.

(132) FIG. **32** shows another embodiment of a sole structure **210** that may be used in place of the sole structure **10** in the article of footwear **12**. The sole structure **210** has the same midsole **14** as described with respect to sole structure **10**, but includes a one-piece outsole **216** that covers the ground-facing surface **40** of the midsole **14** in each of the forefoot region **30**, the midfoot region **28**, and the heel region **26**, and establishes the ground-engaging surface of the sole structure **210**. The one-piece outsole **216** may entirely cover each of these regions so that no portion of the ground-facing surface **40** of the midsole **14** is a ground-engaging surface of the sole structure **210**.

(133) FIGS. **33** and **34** show the truncated peripheral protrusions **44D**. FIGS. **38** and **42** best show the medial-lateral stability afforded by the truncated protrusions **44D** in the heel region **26**. FIG. **35** best shows the arced profile of the sole structure **210**. FIGS. **36-42** show the flaring side walls **56**, **58**. FIG. **39** also shows the medial side wall **56** and the lateral side wall **58** flaring outward from the upper extent **56A**, **58A** to the lower extent **56B**, **58B**, respectively. FIG. **39** shows that the outsole **216** extends to and underlies the medial side edge **56C** and the lateral side edge **58C** of the midsole **14**, terminating at the medial side edge **56C** and the lateral side edge **58C** without extending onto the medial side wall **56** and the lateral side wall **58**.

(134) FIG. **43** shows another embodiment of a sole structure **310** that may be used in place of the sole structure **10** in the article of footwear **12**. The sole structure **310** has the same midsole **14** as described with respect to sole structure **10**, but includes a two-piece outsole **316**. The outsole **316** includes a forefoot outsole element **316A** (also referred to as a first outsole element), and a heel outsole element **316B** (also referred to as a second outsole element). The forefoot outsole element **316A** covers the majority of the ground-facing surface **40** in the forefoot region **30** and has a non-linear rear edge **353**. The heel outsole element **316B** covers the majority of the ground-facing surface **40** of the midsole **14** in the heel region **26** and has a nonlinear front edge **361**. Many of the downwardly-extending protrusions **44** in the midfoot region **28** (only some of which are labelled) are not covered by any portion of the outsole **316** so that the ground-facing surface **40** in that area also serves as the ground-engaging surface of the sole structure **310**.

(135) FIGS. **44** and **45** show the truncated peripheral protrusions **44D**. FIGS. **49** and **53** best show the medial-lateral stability afforded by the truncated protrusions **44D** in the heel region **26**. FIG. **46** best shows the arced profile of the sole structure **310**. FIGS. **47-50** show the flaring side walls **56**,

58.

(136) FIGS. 48-49 show the medial side wall 56 and the lateral side wall 58 flaring outward from the upper extent 56A, 58A to the lower extent 56B, 58B. FIG. 50 shows that the outsole 316 (forefoot outsole element 316A) extends to and underlies the medial side edge 56C and the lateral side edge 58C of the midsole 14, terminating at the medial side edge 56C and the lateral side edge 58C without extending onto the medial side wall 56 and the lateral side wall 58. As shown in FIG. 53, the heel outsole element 316B extends to and underlies the medial side edge 56C and the lateral side edge 58C of the midsole 14, terminating at the medial side edge 56C and the lateral side edge 58C without extending onto the medial side wall 56 and the lateral side wall 58. FIGS. 51 and 52 illustrate that in some portions of the midfoot region 28, the ground-facing surface 40 of the midsole 14 is not covered by the outsole 16 and thus also serves as the ground-contact surface.

(137) The following Clauses provide example configurations of an article of footwear disclosed herein. Clause 1. A sole structure for an article of footwear, the sole structure comprising: a midsole having a ground-facing surface with a forefoot region, a midfoot region, and a heel region, the midsole defining downwardly-extending protrusions at the ground-facing surface distributed over the midfoot region and the heel region, each of the downwardly-extending protrusions having a convex outer surface; and wherein a height of the midsole is greatest at the midfoot region. Clause 2. The sole structure of clause 1, wherein the ground-facing surface of the midsole curves upwardly from the midfoot region to a forward extent of the midsole and from the midfoot region to a rear extent of the midsole, establishing an arced profile of the midsole. Clause 3. The sole structure of clause 1, wherein the ground-facing surface of the forefoot region is relatively flat. Clause 4. The sole structure of clause 1, wherein the midsole is characterized by an absence of downwardly-extending protrusions in the forefoot region. Clause 5. The sole structure of any of clauses 1-4, wherein the midsole includes a rear side wall that flares outward from an upper extent to a lower extent of the rear side wall at a rear of the heel region. Clause 6. The sole structure of any of clauses 1-5, wherein the midsole includes a lateral side wall and a medial side wall each of which has an upper extent and a lower extent and each of which flares outward from the upper extent to the lower extent in the forefoot region. Clause 7. The sole structure of any of clauses 1-6, wherein the downwardly-extending protrusions include forwardmost protrusions each having a front half and a rear half, and with the convex outer surface steeper at the rear half than at the front half. Clause 8. The sole structure of any of clauses 1-7, wherein: the downwardly-extending protrusions include peripheral protrusions in the midfoot region that define a lateral side edge and a medial side edge of the ground-facing surface; and the peripheral protrusions are truncated at the lateral side edge and at the medial side edge such that peaks of the peripheral protrusions lie along the lateral side edge and the medial side edge. Clause 9. The sole structure of any of clauses 1-8, wherein the midsole is a one-piece foam body. Clause 10. The sole structure of any of clauses 1-9, further comprising: an outsole including an outsole element covering the ground-facing surface of the midsole in the forefoot region. Clause 11. The sole structure of clause 10, wherein the outsole element is a first outsole element, and the outsole further includes a second outsole element covering the ground-facing surface of the midsole in the heel region. Clause 12. The sole structure of clause 11, wherein: the outsole further includes a third outsole element covering the ground-facing surface of the midsole in the midfoot region; the first outsole element and the third outsole element are separated by a first gap extending from a medial edge to a lateral edge of the midsole; the midsole includes a first ridge that extends into and at least partially fills the first gap; the second outsole element and the third outsole element separated by a second gap extending from a medial edge to a lateral edge of the midsole; and the midsole includes a second ridge that extends into and at least partially fills the second gap. Clause 13. The sole structure of clause 12, wherein the first ridge and the second ridge are nonlinear. Clause 14. The sole structure of any of clauses 1-9, further comprising: an outsole covering at least a portion of the ground-facing surface of the midsole; and wherein a height of the sole structure is greatest at the midfoot region. Clause 15. The sole structure

of clause 14, wherein the outsole is thicker at peaks of the downwardly-extending protrusions than at the ground-facing surface of the midsole between adjacent peaks of the downwardly-extending protrusions. Clause 16. The sole structure of clause 14 or 15, wherein: the midsole includes a medial side wall having a lower medial side edge and a lateral side wall having a lower lateral side edge; and the outsole extends to and underlies the medial side edge and the lateral side edge, terminating at the medial side edge and the lateral side edge without extending onto the medial side wall and the lateral side wall. Clause 17. The sole structure of any of clauses 14-16, wherein: the outsole includes a forefoot outsole element, a midfoot outsole element, and a heel outsole element, each of the forefoot outsole element, the midfoot outsole element, and the heel outsole element extending from a medial side wall to a lateral side wall of the midsole; a rear edge of the forefoot outsole element is spaced apart from a forward edge of the midfoot outsole element defining a first gap between the rear edge of the forefoot outsole element and the forward edge of the midfoot outsole element; and a rear edge of the midfoot outsole element is spaced apart from a forward edge of the heel outsole element defining a second gap between the rear edge of the midfoot outsole element and the forward edge of the heel outsole element. Clause 18. The sole structure of clause 17, wherein the rear edge of the forefoot outsole element has an irregular shape, and the forward edge of the midfoot outsole element has a complementary irregular shape that tracks the irregular shape of the rear edge of the forefoot outsole element. Clause 19. The sole structure of clause 17 or 18, wherein the forward edge of the heel outsole element has an irregular shape, and the rear edge of the midfoot outsole element has a complementary irregular shape that tracks the irregular shape of the forward edge of the heel outsole element. Clause 20. The sole structure of clause 19, wherein the midsole includes a first ridge that extends into and at least partially fills the first gap, and the midsole includes a second ridge that extends into and at least partially fills the second gap. Clause 21. A sole structure for an article of footwear, the sole structure comprising: a midsole having a ground-facing surface with a forefoot region, a midfoot region, and a heel region, the midsole defining downwardly-extending protrusions at the ground-facing surface distributed over the midfoot region and the heel region and absent from at least a forward half of the forefoot region, each of the downwardly-extending protrusions having a convex outer surface, the downwardly-extending protrusions including: midfoot protrusions in the midfoot region having widths in a transverse direction of the midsole greater than lengths in a longitudinal direction of the midsole, and heel protrusions in the heel region having width-to-length ratios less than width-to-length ratios of the midfoot protrusions. Clause 22. The sole structure of clause 21, wherein a height of the midsole is greater at the midfoot protrusions than at the heel protrusions, and greater at the midfoot protrusions than at the forefoot region. Clause 23. The sole structure of clause 22, wherein the ground-facing surface of the midsole curves upwardly from the midfoot region to a forward extent of the midsole and from the midfoot region to a rear extent of the midsole, establishing an arced profile of the midsole. Clause 24. The sole structure of any of clauses 21-23, wherein the midsole includes a rear side wall that flares outward from an upper extent to a lower extent of the rear side wall at a rear of the heel region. Clause 25. The sole structure of any of clauses 21-23, wherein the midsole includes a lateral side wall and a medial side wall each of which has an upper extent and a lower extent and flares outward from the upper extent to the lower extent in the forefoot region. Clause 26. The sole structure of any of clauses 21-23, wherein the downwardly-extending protrusions include forwardmost protrusions each having a front half and a rear half, and with the convex outer surface steeper at the rear half than at the front half. Clause 27. The sole structure of any of clauses 21-23, wherein the midsole is a foam body and the midfoot protrusions are a ground contact surface of the foam body. Clause 28. The sole structure of any of clauses 12-23, wherein the midsole is a foam body and the heel protrusions are a ground contact surface of the foam body. Clause 29. The sole structure of any of clauses 21-23, further comprising: an outsole element covering the ground-facing surface of the midsole only in the forefoot region. Clause 30. The sole structure of clause 29, wherein forwardmost protrusions of the downwardly-

extending protrusions are rearward of a widest portion of the midsole in the forefoot region, and a majority of the outsole element is forward of the widest portion of the midsole in the forefoot region. Clause 31. The sole structure of any of clauses 21-23, wherein spacing between adjacent ones of the heel protrusions is greater than spacing between adjacent ones of the midfoot protrusions. Clause 32. The sole structure of any of clauses 21-23, wherein: the downwardly-extending protrusions include peripheral protrusions in the midfoot region that define a lateral side edge and a medial side edge of the ground-facing surface; and the peripheral protrusions are truncated at the lateral side edge and at the medial side edge such that peaks of the peripheral protrusions lie along the lateral side edge and the medial side edge. Clause 33. The sole structure of any of clauses 21-23, wherein the midsole is a one-piece foam body. Clause 34. A sole structure for an article of footwear, the sole structure comprising: a midsole having a ground-facing surface with a forefoot region, a midfoot region, and a heel region, the midsole defining downwardly-extending protrusions distributed over the midfoot region and the heel region, and the forefoot region being relatively flat; wherein the downwardly-extending protrusions have convex outer surfaces and transitioning in a forward direction from relatively round to relatively oblong and back to relatively round, each of the relatively oblong downwardly-extending protrusions having a width in a transverse direction of the midsole greater than a length in a longitudinal direction of the midsole, and wherein the relatively oblong downwardly-extending protrusions are taller than the relatively round downwardly-extending protrusions, and the midsole arcs upward from the midfoot region to the heel region and upward from the midfoot region to the forefoot region. Clause 35. The sole structure of clause 34, wherein the downwardly-extending protrusions are a ground contact surface of the midsole. Clause 36. The sole structure of clause 35, further comprising: an outsole element covering the ground-facing surface of the midsole only in the forefoot region. Clause 37. The sole structure of any of clauses 34-36, wherein the midsole includes a rear side wall that flares outward from an upper extent to a lower extent of the rear side wall at a rear of the heel region. Clause 38. The sole structure of any of clauses 34-36, wherein the midsole includes a lateral side wall and a medial side wall each of which has an upper extent and a lower extent and flares outward from the upper extent to the lower extent in the forefoot region. Clause 39. The sole structure of any of clauses 34-36, wherein the downwardly-extending protrusions include forwardmost protrusions each having a front half and a rear half, and with the convex outer surface steeper at the rear half than at the front half. Clause 40. The sole structure of any of clauses 34-36, wherein: the downwardly-extending protrusions include peripheral protrusions in the midfoot region that define a lateral side edge and a medial side edge of the ground-facing surface; and the peripheral protrusions are truncated at the lateral side edge and at the medial side edge such that peaks of the peripheral protrusions lie along the lateral side edge and the medial side edge.

(138) To assist and clarify the description of various embodiments, various terms are defined herein. Unless otherwise indicated, the following definitions apply throughout this specification (including the claims). Additionally, all references referred to are incorporated herein in their entirety.

(139) An “article of footwear”, a “footwear article of manufacture”, and “footwear” may be considered to be both a machine and a manufacture. Assembled, ready to wear footwear articles (e.g., shoes, sandals, boots, etc.), as well as discrete components of footwear articles (such as a midsole, an outsole, an upper component, etc.) prior to final assembly into ready to wear footwear articles, are considered and alternatively referred to herein in either the singular or plural as “article(s) of footwear”.

(140) “A”, “an”, “the”, “at least one”, and “one or more” are used interchangeably to indicate that at least one of the items is present. A plurality of such items may be present unless the context clearly indicates otherwise. All numerical values of parameters (e.g., of quantities or conditions) in this specification, unless otherwise indicated expressly or clearly in view of the context, including the appended claims, are to be understood as being modified in all instances by the term “about”

whether or not “about” actually appears before the numerical value. “About” indicates that the stated numerical value allows some slight imprecision (with some approach to exactness in the value; approximately or reasonably close to the value; nearly). If the imprecision provided by “about” is not otherwise understood in the art with this ordinary meaning, then “about” as used herein indicates at least variations that may arise from ordinary methods of measuring and using such parameters. In addition, a disclosure of a range is to be understood as specifically disclosing all values and further divided ranges within the range.

(141) The terms “comprising”, “including”, and “having” are inclusive and therefore specify the presence of stated features, steps, operations, elements, or components, but do not preclude the presence or addition of one or more other features, steps, operations, elements, or components. Orders of steps, processes, and operations may be altered when possible, and additional or alternative steps may be employed. As used in this specification, the term “or” includes any one and all combinations of the associated listed items. The term “any of” is understood to include any possible combination of referenced items, including “any one of” the referenced items. The term “any of” is understood to include any possible combination of referenced claims of the appended claims, including “any one of” the referenced claims.

(142) For consistency and convenience, directional adjectives may be employed throughout this detailed description corresponding to the illustrated embodiments. Those having ordinary skill in the art will recognize that terms such as “above”, “below”, “upward”, “downward”, “top”, “bottom”, etc., may be used descriptively relative to the figures, without representing limitations on the scope of the invention, as defined by the claims.

(143) The term “longitudinal” refers to a direction extending along a length of a component. For example, a longitudinal direction of a shoe extends between a forefoot region and a heel region of the shoe. The term “forward” or “anterior” is used to refer to the general direction from a heel region toward a forefoot region, and the term “rearward” or “posterior” is used to refer to the opposite direction, i.e., the direction from the forefoot region toward the heel region. In some cases, a component may be identified with a longitudinal axis as well as a forward and rearward longitudinal direction along that axis. The longitudinal direction or axis may also be referred to as an anterior-posterior direction or axis.

(144) The term “transverse” refers to a direction extending along a width of a component. For example, a transverse direction of a shoe extends between a lateral side and a medial side of the shoe. The transverse direction or axis may also be referred to as a lateral direction or axis or a mediolateral direction or axis.

(145) The term “vertical” refers to a direction generally perpendicular to both the lateral and longitudinal directions. For example, in cases where a sole is planted flat on a ground surface, the vertical direction may extend from the ground surface upward. It will be understood that each of these directional adjectives may be applied to individual components of a sole. The term “upward” or “upwards” refers to the vertical direction pointing towards a top of the component, which may include an instep, a fastening region and/or a throat of an upper. The term “downward” or “downwards” refers to the vertical direction pointing opposite the upwards direction, toward the bottom of a component and may generally point towards the bottom of a sole structure of an article of footwear.

(146) While various embodiments have been described, the description is intended to be exemplary, rather than limiting and it will be apparent to those of ordinary skill in the art that many more embodiments and implementations are possible that are within the scope of the embodiments. Any feature of any embodiment may be used in combination with or substituted for any other feature or element in any other embodiment unless specifically restricted. Accordingly, the embodiments are not to be restricted except in light of the attached claims and their equivalents. Also, various modifications and changes may be made within the scope of the attached claims.

(147) While several modes for carrying out the many aspects of the present teachings have been

described in detail, those familiar with the art to which these teachings relate will recognize various alternative aspects for practicing the present teachings that are within the scope of the appended claims. It is intended that all matter contained in the above description or shown in the accompanying drawings shall be interpreted as illustrative and exemplary of the entire range of alternative embodiments that an ordinarily skilled artisan would recognize as implied by, structurally and/or functionally equivalent to, or otherwise rendered obvious based upon the included content, and not as limited solely to those explicitly depicted and/or described embodiments.

Claims

1. A sole structure for an article of footwear, the sole structure comprising: a midsole having a ground-facing surface with a forefoot region, a midfoot region, and a heel region, the midsole defining downwardly-extending protrusions at the ground-facing surface distributed over the midfoot region and the heel region, each of the downwardly-extending protrusions having a convex outer surface; and an outsole covering at least a portion of the ground-facing surface of the midsole; wherein: a height of the sole structure is greatest at the midfoot region; a height of the midsole from a foot-facing surface of the midsole on which a foot rests to the ground-facing surface of the midsole is greatest at the midfoot region such that a height of the midsole in the heel region from the foot-facing surface to the ground-facing surface is less than the height of the midsole in the midfoot region; the outsole includes a forefoot outsole portion, a midfoot outsole portion, and a heel outsole portion, each of the forefoot outsole portion, the midfoot outsole portion, and the heel outsole portion extending from a medial side wall to a lateral side wall of the midsole; a rear edge of the forefoot outsole portion is spaced apart from a forward edge of the midfoot outsole portion defining a first gap between the rear edge of the forefoot outsole portion and the forward edge of the midfoot outsole portion; a rear edge of the midfoot outsole portion is spaced apart from a forward edge of the heel outsole portion defining a second gap between the rear edge of the midfoot outsole portion and the forward edge of the heel outsole portion; wherein the outsole is thicker at peaks of the downwardly-extending protrusions than at the ground-facing surface of the midsole between adjacent peaks of the downwardly-extending protrusions; and wherein the rear edge of the forefoot outsole portion has an irregular shape that is nonlinear from a medial extent of the rear edge of the forefoot outsole portion to a lateral extent of the rear edge of the forefoot outsole portion and is furthest forward between the medial extent of the forefoot outsole portion and the lateral extent of the forefoot outsole portion, and the forward edge of the midfoot outsole portion has a complementary irregular shape that tracks the irregular shape of the rear edge of the forefoot outsole portion such that a width of the first gap is substantially constant.
2. The sole structure of claim 1, wherein the forward edge of the heel outsole portion has an irregular shape, and the rear edge of the midfoot outsole portion has a complementary irregular shape that tracks the irregular shape of the forward edge of the heel outsole portion.
3. The sole structure of claim 1, wherein the ground-facing surface of the midsole in the heel region has a nonlinear profile from a forward extent of the heel region to a rear extent of the midsole that curves upwardly continuously from the forward extent of the heel region to the rear extent of the midsole.
4. The sole structure of claim 3, wherein the ground-facing surface of the midsole curves upwardly from the midfoot region to a forward extent of the midsole establishing an arced profile of the midsole.
5. The sole structure of claim 3, wherein the forefoot region extends from the midfoot region to a forward extent of the midsole, and the midsole is characterized by an absence of downwardly-extending protrusions in a forward half of the forefoot region.
6. The sole structure of claim 3, wherein the midsole includes a rear side wall that flares outward

from an upper extent to a lower extent of the rear side wall at a rear of the heel region.

7. The sole structure of claim 3, wherein the midsole includes a lateral side wall and a medial side wall each of which has an upper extent and a lower extent and each of which flares outward from the upper extent to the lower extent in the forefoot region.

8. The sole structure of claim 3, wherein the downwardly-extending protrusions include forwardmost protrusions each having a front half and a rear half, and with the convex outer surface steeper at the rear half than at the front half.

9. The sole structure of claim 8, wherein the front half extends rearward to a peak and the rear half extends rearward from the peak.

10. The sole structure of claim 3, wherein: the downwardly-extending protrusions include peripheral protrusions in the midfoot region that define a lateral side edge and a medial side edge of the ground-facing surface; and the peripheral protrusions are truncated at the lateral side edge and at the medial side edge such that peaks of the peripheral protrusions lie along the lateral side edge and the medial side edge and the peripheral protrusions slope upward and inward from the peaks at the medial side edge and the lateral side edge.

11. The sole structure of claim 3, wherein the midsole is a one-piece foam body.

12. The sole structure of claim 3, wherein: the midsole includes a medial side wall having a lower medial side edge and a lateral side wall having a lower lateral side edge; and the outsole extends to and underlies the lower medial side edge and the lower lateral side edge, terminating at the lower medial side edge and the lower lateral side edge without extending onto the medial side wall and the lateral side wall.

13. The sole structure of claim 3, wherein the ground-facing surface of the midsole curves continuously from a posterior extent of the heel region to an anterior extent of the forefoot region.
